HEAT-EWS

15-Day Hackathon MVP | Software Requirements Specification | Version 3.0

AI-Assisted Occupational Heat and Environmental Early Warning | Abu Dhabi, UAE

Status: Implementation-ready proposal | **Timebox:** 15 calendar days | **Evidence:** Open environmental data + synthetic workplace aggregates.

Important: This is a prototype decision-support system, not a clinically validated, deployed occupational safety service. All alert thresholds and validation targets below are proposals, not established safety limits.

1. Purpose and challenge alignment

Build an explainable, privacy-preserving early-warning dashboard that combines changing environmental conditions with aggregated workplace health and exposure signals, detects emerging site-level risks, and recommends targeted preventive actions before heat-related incidents escalate. Focus on 3 fictional outdoor construction sites in Abu Dhabi, UAE; use approximate non-identifying site coordinates and Asia/Dubai (UTC+04:00) time zone. Target HSE officers, site operations leads and aggregate public-health viewers.

Challenge requirement	MVP evidence / acceptance demonstration
1. Multi-signal surveillance	Live or cached weather (temperature, humidity) + air-quality PM2.5; independent synthetic site-shift incident / near-miss and absence counts.
2. Explainable anomaly detection	Rolling historical baseline, CUSUM or robust z-score, contribution-by-signal chart, site/time window and confidence flags.
3. Actionable decision support	Watch / Advisory / Action tiers; role-specific playbook, supporting measurements, explicit assumptions and acknowledgement.
4. Privacy-preserving views	Aggregate-only input, site-scoped RBAC, counts 1-10 suppressed, no employee identifiers or employment decisions.
5. Operational validation	Chronological replay of labeled synthetic scenarios; lead time, sensitivity, PPV, false alarms, latency, and missing-data degradation.
6. Health and resilience impact	Simulated prevented-exposure opportunity and incident / absence impact scenarios, explicitly illustrative rather than observed outcomes.

2. Scope and exclusions

2.1 In scope (must deliver)

• Weather and PM2.5 environmental ingestion for selected site coordinates using a public provider; cache responses and preserve timestamps.

• Aggregated workplace CSV import: workers_on_shift, worker_hours, heat_incidents, near_misses, sickness_absence_count, shift_date and site_id.

• Historical/synthetic replay, statistical anomaly detection, a lightweight Logistic Regression next-24-hour simulated incident-episode classifier, risk tiers and 24-72 hour forecast-informed exposure outlook.

• Interactive React dashboard with site comparison, alert history, evidence cards, action tracking, and validation metrics.

• Privacy filtering, site-level authorization, freshness labels, and degraded-input handling.

2.2 Explicitly out of scope

• Hospital/EHR integrations, real clinical patient data, wearables, real worker tracking, diagnosis and regulatory enforcement.

• Deep learning, XGBoost/SHAP, sophisticated WBGT/Liljegren modeling, and claims of clinically calibrated illness probabilities or verified forecast accuracy without real-world labels.

• SMS infrastructure, multi-language support, complex geospatial heatmaps, enterprise SSO, real-time IoT devices and production-grade scale.

• Verified reduction in actual illness or absenteeism; these require a real pilot and longer observation.

3. Stakeholders, roles and access

Role	Allowed actions	Restrictions
HSE Officer	Own-site dashboard, evidence, acknowledge and log preventive action	No individual records; no other sites
District/Public Health Viewer	Aggregated district-level patterns, suppressed site risk tiers	No small cells or employee data
Data Steward / Admin	Register sites, upload simulated aggregates, review data health, configure thresholds, view audit	Cannot bypass privacy suppression in exported views

Demo authentication: predefined role-based test accounts with server-enforced site scopes. Production MFA and identity management are post-MVP tasks. No worker ID, name, contact, or individual health field may enter the schema.

4. System architecture and technology

Frontend: React + Vite + responsive chart library; deploy to Vercel. **Backend:** FastAPI + Pydantic + SQLAlchemy; deploy on a free-tier-compatible Python host, subject to current service limits. **Database:** managed free-tier PostgreSQL (preferred) or hosted SQLite only if durable storage is guaranteed. **Analytics:** pandas/numpy and scikit-learn LogisticRegression for preprocessing, statistical baselines, supervised simulation experiments and metrics. **Data:** Open-Meteo weather and air-quality APIs, with attribution and license checks. **Jobs:** manual refresh + scheduled refresh where supported; cached fallback.

Pipeline: Provider fetch / CSV upload -> validation + aggregation -> storage -> anomaly detector + ML classifier -> safety decision engine + tier cap -> Evidence Card -> React dashboard + action log -> replay metrics.

5. Functional requirements (M = mandatory, S = stretch)

ID	Pri	Requirement
ING-01	M	Fetch current and forecast temperature, relative humidity and PM2.5 per configured site; store provider, event/issue time and arrival time.
ING-02	M	Validate uploaded site-shift aggregates, reject identifiers, negatives, duplicates and impossible ratios; return row errors.
ING-03	M	Mark data stale using configurable source-specific thresholds and show age / last refresh.
DET-01	M	Compute baseline using previous 7-14 comparable site-days or supplied clean simulated history; use only past observations.
DET-02	M	Calculate positive deviations of incidents, near misses and absence per worker-hours or workforce; report baseline and observed values.
DET-03	M	Apply one-sided CUSUM to standardized incident rates OR a robust rolling z-score, using documented tunable parameters.
DET-04	M	Flag environmental anomalies for heat index and PM2.5 relative to historical or reference levels, not arbitrary unlabelled clinical thresholds.
FCT-01	M	Produce 24/48/72 h provider-based environmental outlook; separately predict a labeled next-24 h simulated elevated incident episode using Logistic Regression, with explicit simulation-only limitations.
FUS-01	M	Fuse statistical anomalies, environmental exposures and optional ML evidence with documented, tunable decision rules; do not assert arbitrary weights are validated.

ID	Pri	Requirement
FUS-02	M	Generate Watch, Advisory and Action tiers. Action requires corroboration and adequate freshness; cap tier when key data missing.
FUS-03	M	Deduplicate repeated alerts within configurable cooldown window; record alert time and site.
EXP-01	M	Every alert shows hazard, site, time window, observed vs baseline, contributing signals, missing feeds, freshness, tier logic and assumptions.
ACT-01	M	Each tier includes preventive action, responsible role and review urgency; support acknowledgement and action log.
PRV-01	M	Store only site/day/shift aggregate data; block identifiers and enforce server-side role/site access.
PRV-02	M	Suppress counts 1-10, derived percentages and reconstructable totals in outward-facing views and exports.
VAL-01	M	Replay historical/synthetic cases chronologically with event-time and availability-time cutoffs; avoid future leakage.
VAL-02	M	Report lead time, sensitivity, PPV, false alerts per site-month, data latency, missing-input tests and comparison to naive heat-only baseline.
IMP-01	M	Show clearly labeled scenario-based impact estimates, with explicit assumptions; do not present as actual prevention outcomes.
UI-01	M	Responsive overview, site detail, Evidence Card, validation page, CSV upload and data-quality status.
OPS-01	M	Expose health endpoint, error messages, source attribution, config version and a documented deployment procedure.
NOT-01	S	Send optional email/webhook notification when a new Action tier appears.
MAP-01	S	Display simple site pins if site-level coordinates are available and suitably generalized.

6. Data model and sources

Entity	Minimum fields
sites	site_id, pseudonym, sector, region, latitude, longitude, timezone
environment_records	site_id, valid_at, issued_at, received_at, temperature_c, humidity_pct, pm25, source, is_forecast
workplace_daily	site_id, shift_date, workers_on_shift, worker_hours, heat_incidents, near_misses, sickness_absence_count, received_at, synthetic_flag
risk_evaluations	site_id, evaluated_at, heat_signal, pm25_signal, workplace_signal, score, ml_score, model_version, tier, quality_flags, config_version
alerts / actions	alert_id, site_id, tier, issued_at, evidence_json, acknowledged_at, action_code, action_time, feedback
users / audit	user_id, role, permitted_site_id, action, object_id, timestamp
ml_runs / labels	run_id, model_version, feature_cutoff_at, label_window, simulated_label, predicted_score, split, evaluation_json

Public environmental data may be real, but workplace incident/absence data will be synthetic unless a lawful aggregated partner feed is obtained. Display a visible "Synthetic workplace data" indicator throughout the demo. A missing clinical feed is not silently fabricated or implied to exist.

7. Analytics, alert policy and explanations

7.1 Preprocessing

Normalize to site-local shift day, retain UTC timestamps, deduplicate by site/date/source, range-check temperature/humidity/PM2.5, flag missing values, and preserve source age. Use NOAA Heat Index only when its applicability conditions are met; otherwise show temperature/humidity as screening context. Do not present Heat Index as WBGT or a certified occupational exposure limit.

7.2 Simple anomaly engine

For each site, compare workplace event rates with prior baseline: incident_rate = heat_incidents / max(worker_hours, 1) * 1000; similarly for near misses. For counts, optionally use one-sided CUSUM: S[t] = max(0, S[t-1] + z[t] - k). A flagged deviation must show z, baseline period, k and h. With short history, use a labeled provisional baseline and reduced confidence.

7.3 Lightweight supervised ML: Logistic Regression (mandatory)

Prediction task: for each site at decision time t, classify whether an independently labeled elevated workplace incident episode occurs in the next 24 hours. Candidate features available at t: latest temperature/humidity, applicable heat screening measure, PM2.5, available next-24-hour environmental forecast, lagged incident/near-miss rates, lagged absence rate, worker-hours/workload and seasonal/shift indicators. Do not use future observed incidents, future actual weather, or late-arriving records unavailable at t. Document feature availability and missing-value treatment.

Training: scikit-learn Pipeline with imputation, scaling and LogisticRegression (regularization); fit preprocessors only on training data. Split chronologically by scenario and time, with a 24-hour label-gap/embargo as needed. Independently generate labels from scenario event timelines, NOT from the risk formula or from the model output. Evaluate held-out scenarios, multiple random simulation seeds, and optionally leave-one-site-out. Report AUROC/AUPRC where meaningful, precision, recall, PPV, confusion matrix, Brier score and reliability plots when sample size permits. Coefficients show associations within the simulator, not causal occupational risk factors.

The ML score is an experimental simulator-derived prediction, NOT a validated probability of real worker illness. If independent labels or sufficient data are unavailable, display ML as unvalidated/demo-only and fall back to the anomaly-plus-safety engine. Save model version, cutoff, training configuration and evaluation provenance.

7.4 Risk fusion and alert policy: unvalidated until tested

The previous example score 0.40*heat + 0.20*PM2.5 + 0.40*workplace was an illustrative heuristic, NOT a scientifically validated Abu Dhabi formula. Do not hard-code it as ground truth or train labels from it. Compare (A) heat-only threshold baseline, (B) statistical anomaly/rule baseline, (C) Logistic Regression and (D) hybrid ML + rule-gated decisions on held-out scenarios. Choose fusion weights or thresholds on development scenarios only, freeze before test, and publish sensitivity/PPV/false alarms/lead time and ablations. No claim of externally validated performance.

Provisional tiers: Watch for environmental concern or unusual early evidence; Advisory for stronger/multiple signals; Action only with adequate freshness, independent workplace corroboration and repeated evidence. The model score alone must never issue an Action. Safety rules may prompt heat precautions regardless of ML score. Missing workplace feed: maximum Advisory; missing environmental feed: no new Action; all feeds stale: Degraded/Unavailable, never Safe. Preserve separate risk and data-confidence indicators.

7.5 Abu Dhabi occupational heat context and prevention

Use Asia/Dubai local time, Abu Dhabi pilot coordinates, and applicable UAE occupational heat protection context. UAE midday-break rules commonly restrict outdoor work in direct sun/open areas from 12:30 to 15:00 during 15 June-15 September, subject to exemptions and current official requirements; verify against UAE MOHRE and Abu Dhabi authorities before release. Provide compliance reminders as informational safeguards, not automated legal determinations. Do not infer WBGT from Heat Index or claim a certified exposure limit.

Missing workplace feed: maximum Advisory; missing environmental feed: no new Action; all feeds stale: show Degraded/Unavailable, not a reassuring green Safe state. Keep separate data confidence from health risk. Do not multiply risk by geographic confidence and mistakenly hide serious hazards.

7.4 Prevention playbook

Tier	Recommended response	Owner
Watch	Check water, shade and forecast; verify site conditions; increase monitoring.	HSE Officer
Advisory	Plan extra shaded breaks, hydration reminders, task re-timing and supervisor briefing.	HSE Officer + Operations Lead
Action	Urgent review of heavy-work timing, cooling access and work-rest arrangements; consider pausing high-exposure tasks subject to qualified local safety policy.	Operations Lead / HSE Officer

Actions are recommendations for human review. The app must never automatically change shifts or issue medical advice. Evidence Cards must distinguish observed measurements, model assumptions, forecasts and synthetic observations.

8. Privacy and security requirements

• Reject worker/patient identifiers at upload; no individual-level tables or drilldowns.

• Apply small-cell suppression: counts 1-10 displayed as "<11"; suppress related percentages and complementary totals when reconstruction is possible.

• Do not publish site-specific small cohorts as exact counts; aggregate across sites or time windows if necessary.

• Server-side authorization, password hashing, HTTPS, environment-based secrets, least-privilege roles and audit trail.

• Explicit prohibition on employee scoring, disciplinary use or individual employment decisions.

• No real personal health data without separate governance, consent/legal basis and privacy review.

9. Validation and measurable acceptance

Build labeled synthetic scenarios with controlled onset and event timing. Freeze thresholds on a development set, then test on a separate held-out set. Run a heat-only naive baseline and a multi-signal version on the same cases. These results demonstrate simulator performance, not clinical efficacy.

Scenario	Expected demonstration
S1: Heat rise + incident spike	Early multi-signal alert, evidence and action
S2: PM2.5 rise + absence/near misses	Combined environmental and workplace warning
S3: Hot weather, no workplace change	Avoid automatic Action; assess false alarms
S4: Workplace incident rise, mild weather	Detect independent workplace anomaly; investigate
S5: Missing workplace CSV	Tier cap, missing-data banner, metric degradation
S6: Stale environmental provider	Cached last-known data labeled stale; no unsupported new Action
S7: Small cohort / counts 1-10	Suppression and access controls enforced
S8: Late-arriving incident report	Replay excludes information not yet received

9.1 Metrics

• Lead time (hours) = simulated incident onset minus first eligible alert timestamp; negative values indicate late warning.

• Sensitivity = TP / (TP + FN); PPV = TP / (TP + FP); define TP matching window before evaluation.

• False alarms = unmatched alerts per site per 30 simulated days; also report alerts per site-month.

• Latency = received_at minus valid_at; report median and 95th percentile by stream.

• Missing-input behavior = change in sensitivity/PPV/false alarms after removing each signal and under late arrivals.

• Impact proxy = number of correctly warned high-exposure site-days with sufficient intervention lead time; illustrative prevented incident range must state assumed effect size.

9.2 MVP acceptance gates (engineering targets, not health-performance promises)

• All six challenge dimensions demonstrable in a single end-to-end walkthrough.

• At least 3 Abu Dhabi sites, 60+ simulated days where feasible, 8 scenario types, a held-out ML evaluation and 24/48/72-hour environmental outlook.

• No individual-level fields accepted; all privacy suppression and role-isolation tests pass.

• Every displayed alert has a complete Evidence Card and named response owner.

• Metrics computed automatically and reproducibly; no invented accuracy or lead-time claims.

• Compare heat-only, statistical anomaly, Logistic Regression and hybrid approaches; report failures, calibration limits and simulator-only conclusions.

• Hosted frontend/backend demo works with seeded data even if external API temporarily fails.

10. Non-functional requirements

ID	Target
NFR-01	Dashboard loads within ~5 seconds on normal broadband for seeded demo data (proposed).
NFR-02	A new CSV import recomputes risk within 60 seconds for 5 demo sites (proposed).
NFR-03	Every environmental data point shows source, timestamp, freshness and synthetic/real provenance.
NFR-04	Deterministic alert output for identical inputs and configuration.
NFR-05	Clear empty/error states and mobile-responsive layout.
NFR-06	System remains demoable using cached API samples and seed database during provider downtime.

11. Dashboard pages and key flows

• Overview: region/site selector, current tier, risk components, latest forecast, stale-data indicator, open alerts.

• Site details: time-series chart of heat, PM2.5 and aggregated workplace signals; baseline overlays.

• Evidence Card: why flagged, when, data used, missing inputs, tier cap, action and responsible role.

• Action tracker: acknowledge, mark reviewed, select action, timestamp and feedback.

• Validation lab: scenario selection, chronological replay, baseline comparison, metrics and ablation table.

• Admin/import: seed data, CSV validation errors, site settings, refresh status and source attribution.

12. 15-day execution plan and priorities

Day	Build / deliverable	Exit gate
1	Freeze pilot sector, 3 Abu Dhabi sites, schemas, thresholds and scenario labels.	SRS + repository + sample CSV
2	FastAPI + PostgreSQL models, seed generator and auth roles.	DB seeded; API health works
3	Weather / PM2.5 provider adapter and caching.	Real API data stored
4	CSV upload, validation, privacy suppression.	Invalid files rejected
5	Heat screening + 24/48/72-hour environmental outlook.	Forecast cards visible
6	Rolling anomaly baseline + feature pipeline and labels.	No leakage; spike detected
7	Train Logistic Regression; build rule-gated fusion and tiers.	Held-out ML report + alert tests
8	Evaluate ML versus baselines; Evidence Card and playbook.	Model comparison + evidence
9	React overview and site-detail charts.	Usable site dashboard
10	Role views, actions, acknowledgements.	Permissions tested
11	Replay harness, scenarios and naive comparator.	All 8 cases executable
12	Replay metrics, ML calibration and missing-stream ablations.	Results reproducible
13	Deploy frontend/backend/database; cache fallback.	Public demo URL works

Day	Build / deliverable	Exit gate
14	End-to-end QA, privacy tests, bug fixes.	Acceptance checklist complete
15	Record demo, write results/limitations, final rehearsal.	Submission-ready package

Critical path: data ingestion -> risk engine -> Evidence Card -> validation. If behind schedule, drop maps, email, custom admin editing and decorative UI first. Never drop missing-data behavior, privacy, evidence cards or validation.

13. Risk register

Risk	Mitigation
No workplace partner data	Use visibly labeled synthetic aggregates; no live clinical claims.
Provider API downtime	Cache responses and ship local sample snapshots.
Free hosting sleeps / quotas	Test deployment by Day 13; provide offline replay.
Too many false alerts	Tune on development scenarios; cooldown and independent corroboration.
Geolocation mismatch	Use approximate fictional Abu Dhabi site coordinates; show provenance and coverage limitations.
Synthetic ML generalization	Never claim real clinical prediction; compare multiple independent simulated scenarios and disclose uncertainty.
Unvalidated risk weights	Treat previous 40/20/40 formula as illustrative; tune and freeze rules on development data, evaluate on held-out scenarios.
Regulatory interpretation	Verify UAE MOHRE midday-break guidance and Abu Dhabi safety requirements with official sources; do not automate legal compliance judgments.
Small-count privacy leakage	Suppression and complementary suppression tests.
15-day time overrun	Strict Must/Should separation; feature freeze after Day 10.

14. Definition of done and submission evidence

• Working React web UI, FastAPI endpoints and persistent database with synthetic demo records.

• At least one real environmental provider, plus simulated workplace CSV feed.

• Eight reproducible scenarios, Logistic Regression training/evaluation, automatic metrics, baseline-vs-ML-vs-hybrid comparison and missing-data ablation.

• Alert evidence card, proportionate prevention playbook, responsible role and action log.

• Role-based privacy demonstration, source attribution and explicit data/validation limitations.

• README, API documentation, architecture diagram, SRS, test results, and a 3-5 minute demo script.

15. Assumptions, limitations and next phase

Assumptions: a pilot region is selected, weather/air-quality data cover its approximate coordinates, and simulated workplace incidents reflect plausible but invented patterns. Environmental forecasts are not a validated prediction of illness. Simulation-derived sensitivity and PPV cannot establish real-world utility or generalizability. Post-hackathon work: obtain lawful real aggregated data, verify exposure thresholds with occupational experts, calibrate forecasts, conduct a shadow-mode pilot and independently review privacy and operational safety.

16. Abu Dhabi data, legal and model references

Official references to verify during implementation: UAE Ministry of Human Resources and Emiratisation (MOHRE) occupational midday-break rules (mohre.gov.ae); Abu Dhabi Public Health Centre Safety in Heat programme (adphc.gov.ae); Open-Meteo Weather and Air Quality API documentation (open-meteo.com); scikit-learn LogisticRegression and model evaluation documentation (scikit-learn.org). API coverage, attribution, local regulatory exemptions and applicable dates must be checked before demo claims. Historical weather and forecast issue-time availability must be distinguished.

17. Source baseline

Adapted from user-provided HEAT-EWS SRS v1.0 (31 October 2026), especially its scope, multi-stream architecture, privacy rules, Evidence Cards, operational validation and compressed implementation path. Existing design references include NIOSH occupational heat criteria, Open-Meteo API documentation, ERA5-Land and statistical surveillance literature. This MVP combines a transparent statistical baseline, an experimental scikit-learn Logistic Regression classifier and provider forecast-based outlook; source citations and legal/licensing requirements should be rechecked before a real pilot.